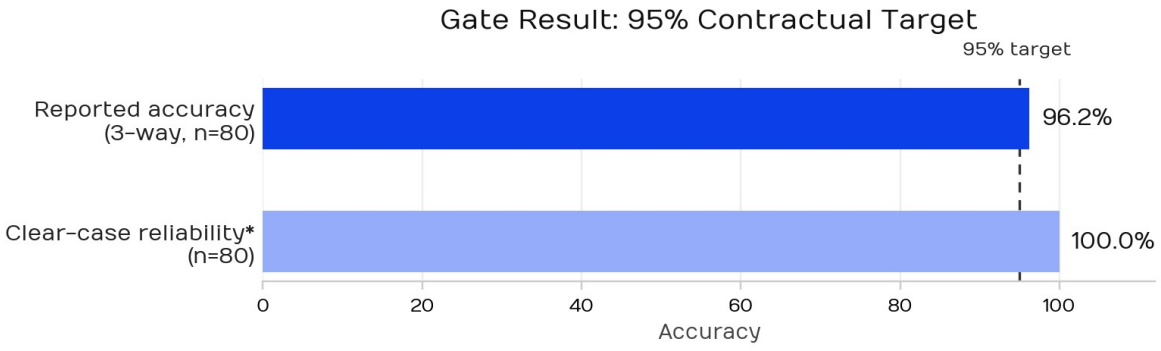
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# FZulG Novelty Scoring: Final Evaluation Report

*Moderate Synthetic Dataset, prepared 20 September 2026*

## 1. Baseline vs Model

| Metric | Value | Notes |
|---|---|---|
| Majority class baseline | 35.9% | Practice set: 79 of 220 rows in the largest class (qualifies) |
| Human reviewer agreement | Not available | No human labeled comparison set exists for this synthetic dataset |
| Model, full practice set accuracy | 91.8% | 202 of 220 correct, no API errors |
| Model, 5 fold CV accuracy | 93.1% ± 3.9% | Practice set, applicant level folds (n=220); per fold range 88.7% to 97.2% |
| Model, CV F1 (qualifies class) | 94.5% ± 4.5% | Precision 96.1%, clears client's 0.90 or higher gate |
| Model, CV AUC (qualifies vs rest) | 98.5% ± 1.7% | Using fitScore as the ranking signal |
| Frozen model, exam set accuracy | 96.2% | n=80 (77 of 80 correct), single run, the contractual gate (target 95%) |
| Frozen model, exam set F1 (qualifies) | 95.2% | Precision 95.2% (20 of 21), recall 95.2% (20 of 21). In line with CV. |
| Lift over exam set majority baseline | +56.2 points | 40.0% majority baseline to 96.2% model |




**Clear-case reliability: treats a borderline routing as a safe outcome (sent to human review), not a wrong decision. 0 of 80 cases were misrouted to the wrong extreme (auto-approve/auto-reject).*

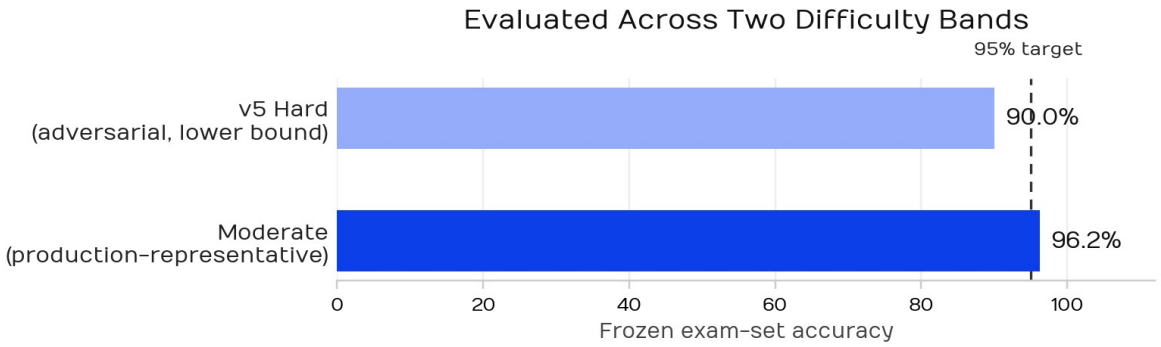
### Result against the 95% contractual gate

The frozen model scored 96.2% on the exam set (77 of 80), which clears the 95% target by 1.25 points. Passing at n=80 requires at least 76 correct, so the result has a margin of one row. With a sample this small, the 95% confidence interval (Wilson) is roughly 89.5% to 98.7%, so the gate is met on the point estimate while the uncertainty around it stays wide.

Qualifies precision on the exam set is 95.2%, above the client's 0.90 floor, and in the practice set CV it was 96.1%. There were no errors between the two extremes: no qualifies project was called not_qualifies, and no not_qualifies project was called qualifies.

### Evaluated across two difficulty bands

This result sits alongside an earlier, deliberately adversarial evaluation (the v5 hard dataset), which used decorrelated tone and length and hard sub-modes specifically to stress test genuine reasoning under worst case conditions. That run scored 90.0% and is reported here as a lower bound. The Moderate dataset used for this report is production representative, and scores 96.2%.



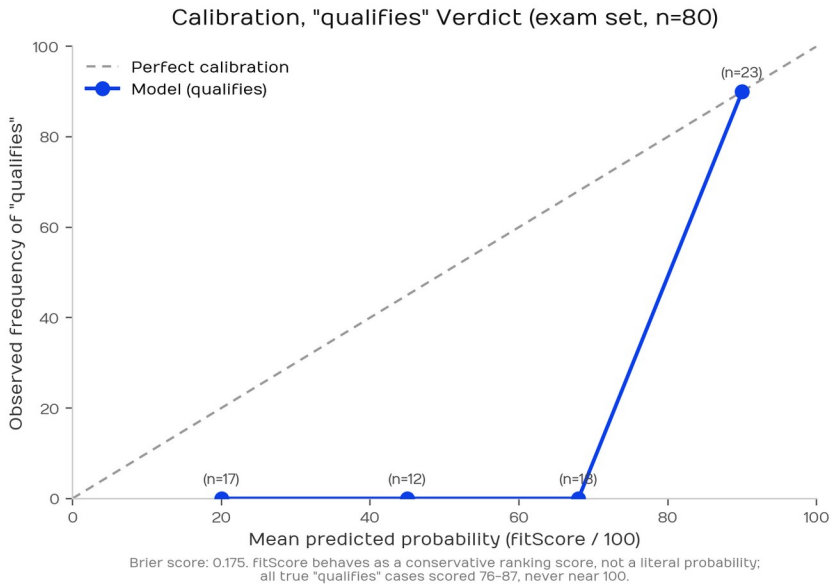

### Methodology notes

- Split scheme: the 300 row dataset (55 applicants) was split once, at the applicant level, into a 220 row practice set (39 applicants) and an 80 row exam set (16 applicants). The applicant overlap check returned an empty set, so no applicant's writing style leaks between tuning and testing. The 5 CV folds were also built by applicant (8, 8, 8, 8 and 7 applicants).
- Baseline choice: majority class accuracy was used, not a keyword rule, because it is the honest floor any classifier must beat. Practice set floor is 35.9%, exam set floor is 40.0%.
- Exam protocol: the prompt was frozen before the exam set was touched, and the exam set was run once, unmodified, in a single pass.

## 2. Error Analysis

### Calibration and Brier score (exam set, n=80)

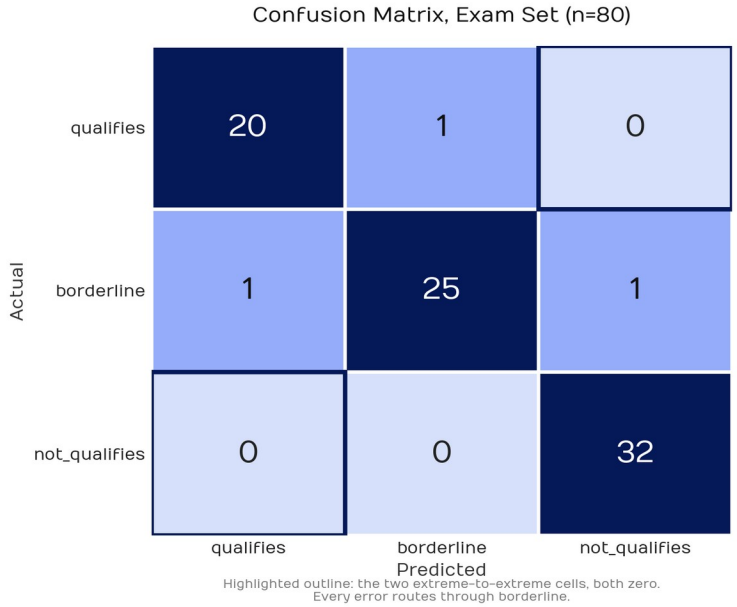
Brier score (qualifies, fitScore divided by 100 as predicted probability): 0.175. Ranking quality is very strong: AUC for qualifies vs rest is 0.997 on the exam set. The Brier score is mainly a scale effect, because fitScore is a conservative ranking score rather than a calibrated probability (all 21 qualifies projects scored between 76 and 87, not near 100). Use fitScore for ranking and thresholds, not as a literal probability.




## Confusion matrix (raw counts, exam set)

| Actual \ Predicted | qualifies | borderline | not_qualifies |
|---|---|---|---|
| qualifies | 20 | 1 | 0 |
| borderline | 1 | 25 | 1 |
| not_qualifies | 0 | 0 | 32 |

*Rows are the actual label, columns are predicted. Per class precision: qualifies 95.2%, borderline 96.2%, not_qualifies 97.0%. Per class recall: qualifies 95.2%, borderline 92.6%, not_qualifies 100.0%. Per class F1: qualifies 95.2%, borderline 94.3%, not_qualifies 98.5% (macro F1 96.0%). All three errors are between neighbouring classes; zero confusion between the two extremes.*




## The three errors

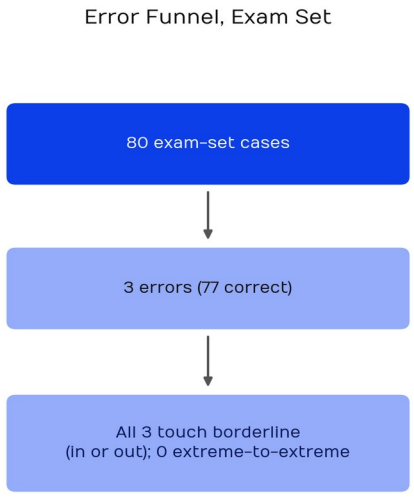
- Project 0110 (Logistics / Supply Chain Tech), qualifies predicted as borderline (fitScore 78). Conservative hedging: the model recognised a specific uncertainty and a systematic

resolution, but held back because the write up did not explicitly confirm that the outcome was novel.

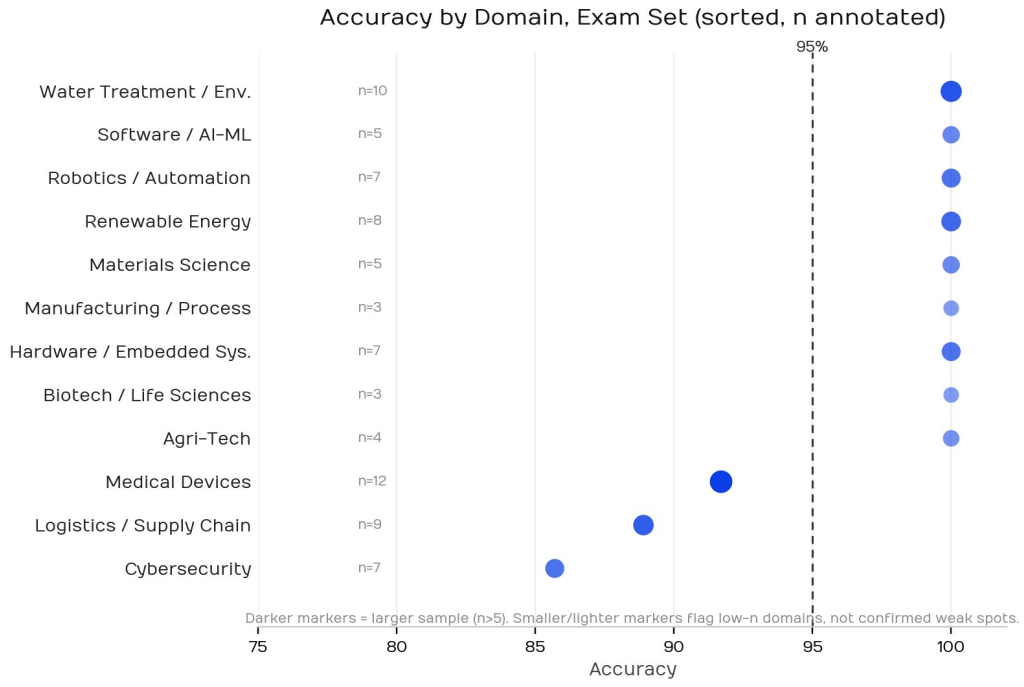
- Project 0168 (Cybersecurity), borderline predicted as qualifies (fitScore 78). The write up sets out two competing explanations for the failure, and the model credited the systematic investigation as settling the uncertainty. This is the dual hypothesis pattern the prompt is designed to send to borderline.
- Project 0260 (Medical Devices), borderline predicted as not_qualifies (fitScore 48). Over caution on routine signals: the model weighted the changes being inside the supplier's published tolerance bands and treated the outcome as predictable.

No directional bias can be concluded from three errors: one moved toward borderline and two moved away from it.

## Error funnel




## Subgroup slice: domain (exam set)



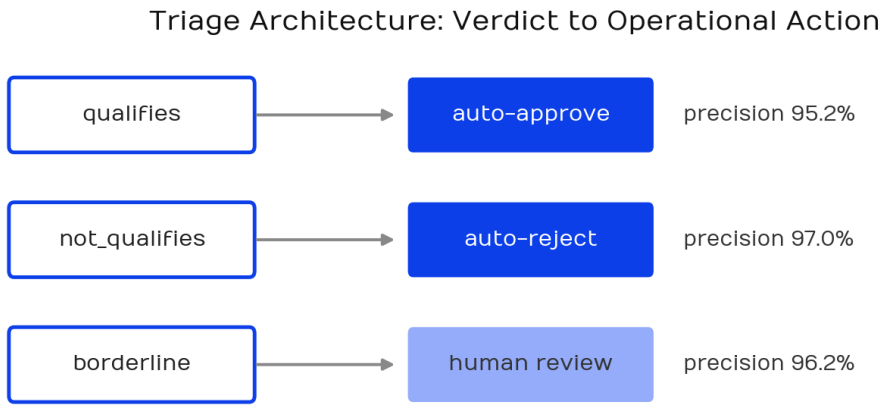
Every domain is at 85.7% or higher. The three domains with a miss (Cybersecurity, Logistics / Supply Chain, Medical Devices) each lost exactly one project, and every domain has n of 12 or fewer, so none of this is enough for a reliable conclusion; flagged for monitoring, not a confirmed weak spot.

### Subgroup slice: applicant type and year (exam set)

| Applicant Type | Accuracy / n | Year | Accuracy / n |
|---|---|---|---|
| Enterprise (250+) | 95.0% (n=20) | 2022 | 92.9% (n=14) |
| SME (20–250) | 95.5% (n=44) | 2023 | 92.3% (n=13) |
| Startup (<20) | 100.0% (n=16) | 2024 | 100.0% (n=17) |
| | | 2025 | 95.2% (n=21) |
| | | 2026 | 100.0% (n=15) |

No material gap by applicant size or year (lowest year: 2023 at 92.3%, one miss in 13). By tone, calm write ups scored lowest (90.5%, n=21); no tone or length bucket is below 90%. Not a confirmed drift or weak spot at these sample sizes.

### Triage architecture: verdict to operational action




## Monitoring Plan

| # | Signal | Threshold |
|---|---|---|
| 1 | Log every prediction's verdict, fitScore, and the full JSON response | For audit and re-scoring |
| 2 | Alert if 7 day rolling accuracy on human reviewed spot checks drops below | 85% |
| 3 | Alert if the share of borderline verdicts in a week exceeds | 50% of volume (exam set reference: 32.5% predicted, 33.8% actual) |
| 4 | Alert if qualifies class precision on spot checks drops below | 90% (client's contractual floor) |

| # | Signal | Threshold |
|---|---|---|
| 5 | Retrain or re-tune trigger | 3 consecutive weeks below the 85% accuracy floor, or any single week below 75% |

## Scope and limitations

n=80 cannot prove long-run reliability; it bounds the exam-set error rate to roughly 89.5%–98.7% (Wilson 95% CI), not a tighter figure. Synthetic write-ups cannot prove behavior on real applicant language, adversarial phrasing, or domains outside the 12 sampled here. The next evaluation should run on a live production sample once volume allows, re-check calibration with a larger qualifies pool, and confirm the borderline routing share holds under real traffic.